Instructor Manual

Competitive Data-Center Connection Planning Game

Power Systems Course

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Instructor purpose. This manual explains how to configure, validate, evaluate, audit, and discuss the competitive data-center connection-planning game. It is the operational companion to the student assignment handout.

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1 Instructor overview

Each group represents a data-center operator. A group chooses connection sites, site capacities, optional private generation, and optional private storage. The instructor collects one canonical site-level CSV per group and evaluates the submissions jointly with a fixed-load DC optimal power flow (`DCOPF`). Groups are ranked by their final score.

Students are explicitly encouraged to use AI tools, including ChatGPT, Codex, Claude Code, GitHub Copilot, and similar systems. The instructor should grade the submitted artifact and the students' engineering explanation, not whether students manually typed every line of code. Students remain responsible for validating generated code and understanding their plan.

The recommended production workflow is:

ACTIVSg2000 public model
⇓
Validated half-year scenario and hourly economic artifacts
⇓
Student CSV collection and format validation
⇓
Joint DCOPF evaluation and fair continuous projection
⇓
Scores, diagnostics, and protected classroom reporting

2 Production mental model

The release uses the following distinction:

- **Public dataset:** the synthetic Texas ACTIVSg2000 topology, buses, branches, generators, status values, PMIN, PMAX, and marginal costs.
- **Derived scenario:** deterministic bus-level hourly load profiles over six half-year blocks.
- **Instructional assumptions:** student-owned assets, IT value, capital costs, storage costs, DCOPF settlement, scoring, and fair projection.

The active scenario package contains 2,000 buses, 3,206 branches, and 544 generator records. ACTIVSg2000 does not provide technology labels or hourly availability fields. Therefore the active release uses each active generator's dataset status, PMIN, and PMAX in every hour; it does not apply synthetic wind/solar derating. The package is not historical ERCOT telemetry, an official ERCOT planning model, or a forecast.

Team-built resources follow a separate convention. A team's `bus_id` must refer to a public bus in the selected scenario block. The canonical CSV submits generator and storage capacities directly; it does not select an existing public generator or storage device by ID. Every team-built generator uses the same convex piecewise-linear operating curve at every bus: 25% of capacity at \$60/MWh, the next 50% at \$100/MWh, and the final 25% at \$160/MWh. Students submit capacities; the evaluator determines hourly dispatch. Team-built generators and storage serve only their submitting team.

3 Released scenario blocks

Each round covers one half-year with a one-hour time step.

Year	Round	Evaluator hours	CSV rows
2028	January–June	4,368	one row per site
2028	July–December	4,416	one row per site
2027 or 2029	January–June	4,344	one row per site
2027 or 2029	July–December	4,416	one row per site

The released artifacts are under `data/processed/activsg2000_scenarios_v1/<year>/<block>`. The root and block manifests record validation status, scenario hashes, and artifact metadata. A block must report `validation_status: passed` before it is used for a classroom round.

4 Student submission contract

The active CSV header is exactly seven columns:

```
team_id,site_id,bus_id,connection_capacity_mw,generator_capacity_mw,
    storage_power_mw,storage_energy_mwh
```

The file has one data row per site and no hourly horizon rows. It has no `period`, hourly `load_mw`, `submission_id`, `generator_id`, or `storage_id` columns. `site_id` is a team-defined site label; `bus_id` must come from the selected block's bus catalog. The evaluator uses the submitted site capacities with the released hourly scenario and applies the common team-built generator curve described above. Generator and storage assets are private to the submitting team and cannot serve another team.

Example:

```
team_id,site_id,bus_id,connection_capacity_mw,generator_capacity_mw,
    storage_power_mw,storage_energy_mwh
team-blue,site-north,1001,40,20,10,40
team-blue,site-south,1002,15,0,0,0
```

The retired ten-column hourly format is historical compatibility input only. Do not distribute it as a current student template.

Storage uses MW power capacity and MWh energy capacity. With ideal charging and discharging efficiency, the evaluator enforces power and energy bounds, no simultaneous charge/discharge, discharge no greater than the stored energy, an initial state of charge of at least $0.5B$, and a terminal state of charge of at least $0.5B$.

5 Course operation

Before the first round, provide students with the student handout, the active scenario block, the CSV checker, and the bus catalog. A practical format is one submission per group per announced

round. The instructor should record the selected year/block, deadline, submission directory, rule-file version, and a hash of the final input directory.

Recommended sequence:

1. Explain the data/model distinction and the CSV contract.
2. Give groups time to explore candidate buses and resource portfolios.
3. Require each group to run the checker before submission.
4. Freeze the submission directory at the deadline.
5. Run the batch evaluator once on the frozen directory.
6. Release scores and aggregate diagnostics without exposing competitors' private plans.

6 Instructor command cookbook

Check one submission:

```
ercot-game-check-csv --file path/to/team.csv
```

Evaluate a frozen round:

```
ercot-game-batch run \
  --submissions-dir work/round_01/submissions \
  --artifact-dir data/processed/activsg2000_scenarios_v1/2028/jan_jun \
  --rules data/teaching/dispatch_scoring_rules_jan_jun_v1.json \
  --year 2028 --round jan_jun \
  --submission-number 2 \
  --workers 8 --projection-iterations 10 \
  --output-dir work/round_01/results
```

The submission number is evaluator-side metadata, not a CSV column. The CLI accepts `1` or `2` and defaults explicitly to `2`. Use `-submission-number 1` for a practice run: it excludes energy-purchase cost while retaining the other active scoring terms. Use `-submission-number 2` for the scored run, which includes locational energy-purchase cost from the joint dispatch.

Validate the complete scenario package:

```
PYTHONPATH=.:src python3 tools/activsg2000/validate_scenario_package.py \
  --package data/processed/activsg2000_scenarios_v1 --workers 6
```

The economic profiles are generated by the module invocation below. The module form is intentional because the builder uses package-relative imports:

```
PYTHONPATH=.:src python3 -m tools.activsg2000.build_economic_profiles \
  --artifact-dir data/processed/activsg2000_scenarios_v1/2028/jan_jun \
  --output-dir data/processed/activsg2000_scenarios_v1/2028/jan_jun
```

7 Evaluation and joint infeasibility

Malformed files are rejected independently so that valid teams remain eligible for evaluation. A valid plan may be feasible alone but infeasible when all teams are dispatched together. In that case, the evaluator computes the largest common $\alpha \in [0, 1]$ such that every valid team's load multiplied by α can be served jointly. This is the fair continuous projection. Submitted asset costs remain part of the score, while served-load value and energy costs reflect the projected load. The output records the projection scope and fraction.

If a common projected dispatch cannot be found, the evaluator records deterministic leave-one-team-out and individual fallback attempts where possible. This prevents one invalid or excessively large submission from silently erasing the scores of otherwise valid teams.

8 Scoring model

At a high level, a team's score is:

$$\begin{aligned}\text{IT value} &- \text{energy purchase cost} - \text{generator operating cost} - \text{generator capital cost} \\ &- \text{storage capital cost} - \text{site cost}.\end{aligned}$$

The current release uses IT value of \$1500/MWh, a team-built generator curve of \$60/\$100/\$160 per MWh at 25/50/25%, site cost of \$20m/MW, generator capital cost of \$1.3m/MW, and storage capital costs of \$350,000/MW plus \$200,000/MWh. A four-hour storage device therefore has a \$1.15m/MW capital equivalent, and storage operating cost is not charged.

Energy purchase cost uses the validated baseline DCOPF LMP at the selected bus and hour. The six released LMP artifacts are stored as deterministic NPZ matrices with one row per hour and one column per bus. Raw balance duals are retained for audit, while nonnegative LMPs are used for scoring. IT value is a complete full-horizon vector supplied in the corresponding economic-profile JSON. These prices are instructional model outputs, not official ERCOT market prices. Submission 1 is the practice mode and sets energy-purchase cost to zero; Submission 2 is the scored mode and includes the locational energy purchase term. The submission number is selected on the instructor CLI and is never represented by an extra CSV column.

9 Concrete multi-team example

The complete example package is in `docs/instructor_manual/examples/multi_team_round/`. It is retained as a historical long-form compatibility package with matching results. It is not a current student CSV template; current submissions use the canonical one-row-per-site contract above. The full raw evaluation record is not checked into the repository because it is very large.

9.1 Single-team CSV excerpt

Each student group submits one CSV file. The current format contains one row per site and no hourly horizon rows:

```
team_id,site_id,bus_id,connection_capacity_mw,generator_capacity_mw,
   storage_power_mw,storage_energy_mwh
team-blue,site-north,1001,40,20,10,40
team-blue,site-south,1002,15,0,0,0
```

The public `bus_id` values must come from the current 2,000-bus artifact; `site_id` is a team-defined label. The checked-in CSVs in the historical example directory retain the retired hourly format solely to preserve a reproducible old result and must be labeled legacy whenever shown.

9.2 Batch collection example

For a round with four groups, the instructor collects four separate files, for example `team-01.csv`, `team-02.csv`, `team-03.csv`, and `team-04.csv`. The batch evaluator reads the directory as a joint submission, but no CSV should contain rows for multiple teams. The leaderboard below is therefore a multi-team result report, not a multi-team submission file.

9.3 Example score report

The following table is populated from the exact generated leaderboard and is not a hand-selected score illustration. This is a historical compatibility result: its checked-in rules, values, and cost fields predate the current release and must not be used as a calibration target for a new round.

Rank	Team	Served load (MWh)	IT value ($)	Energy cost ($)	Asset/storage cost ($)	Score ($)
1	team-03	41,318.88	5,164,859.70	724,983.76	50,000.00	4,389,875.94
2	team-01	41,599.96	5,199,994.80	467,665.19	567,465.38	4,164,864.23
3	team-04	26,779.33	3,347,416.80	301,047.10	388,716.50	2,657,653.19
4	team-02	40,279.73	5,034,966.30	706,971.84	1,945,424.52	2,382,569.94

Asset/storage cost combines site cost, generator capital cost, generator operating cost, storage capital cost, and storage operating cost. Exact component fields are preserved in `leaderboard.csv`; all four submissions were valid and feasible in the joint run.

The decomposition should be read together with the compact `evaluation_summary.json` and the exact `leaderboard.csv`. In a live instructor run, retain the complete raw evaluation record separately. In particular, a large served-load value can improve IT revenue while increasing locational energy cost; storage can reduce or shift energy purchases but adds power-capacity and energy-capacity costs.

9.4 Artifacts to inspect

- `leaderboard.csv`: rank, feasibility, score, and major cost/value fields;
- `submission_manifest.csv`: input validity and SHA-256 hashes;
- `evaluation_summary.json`: compact run metadata, accepted teams, projection outcome, and per-team scores;
- `diagnostic_team_01.json`: one representative team-level diagnostic record;

- `dispatch_period_0.json`: one representative dispatch record;
- `rules/dispatch_scoring_rules_jan_jun_v1.json`: the exact rule file used by the run;
- `evaluation_sha256.txt`: hash of the evaluation record.

10 Privacy and classroom reporting

Keep raw CSVs, per-team diagnostics, bus selections, generator capacities, storage capacities, and hourly profiles instructor-only. Student-facing results may include each team's own score, rank, feasibility status, aggregate served load, common projection fraction, and anonymized network-level summaries. Avoid publishing a map that lets one group reconstruct another group's exact sites or resource portfolio. When showing a network figure, aggregate loads by region or use anonymized selected-resource markers.

11 Reproducibility and release audit

Before release, verify:

- all six scenario blocks report 26,304 total passed hourly states;
- every economic-profile NPZ hash matches its manifest;
- the chosen rule file and scenario block match the round announcement;
- a dry-run CSV passes the checker and batch evaluator;
- the output directory contains the leaderboard, manifest, evaluation record, diagnostics, and hashes;
- the frozen submission directory is archived before any feedback is released.

12 Troubleshooting

Malformed CSV
: Run the checker and return the error to the CSV-format section of the student handout. Do not repair a student's file silently.

Unknown bus
: Confirm that the `bus_id` comes from the selected half-year artifact, not from a retired prototype.

Duplicate site
: Ensure that every canonical data row has a distinct `site_id`.

Joint infeasibility
: Inspect projection metadata and binding diagnostics. A valid team may still receive a projected score.

Missing economic profile

Confirm that the selected block contains `economic_profiles.npz`, `economic_profile.json`, and its manifest, and that the manifest hash matches.

13 Instructor checklist

1. Select the year, round, rule file, and deadline.
2. Publish the student handout and checker.
3. Freeze and hash the submission directory.
4. Run format validation and the full batch evaluation.
5. Inspect valid/invalid counts, projection scope, and diagnostics.
6. Release protected summaries and retain the complete instructor archive.

References